

The figure displays a heatmap comparing the performance of 12 machine learning models across six sample sizes (N=100, N=250, N=500, N=1000, N=2000, N=5000) for two types of variables: Relevant variables and Irrelevant variables. The color scale ranges from 0.00 (light blue) to 2.00 (dark blue).

Relevant variables

Model	N=100	N=250	N=500	N=1000	N=2000	N=5000
lr	1.465 (0.204)	1.949 (0.156)	1.997 (0.084)	1.992 (0.044)	1.99 (0.033)	2.001 (0.021)
lasso	0.196 (0.171)	0.145 (0.086)	0.122 (0.064)	0.088 (0.05)	0.067 (0.041)	0.044 (0.024)
dt	0.492 (0.356)	0.619 (0.213)	0.714 (0.133)	0.845 (0.136)	0.949 (0.107)	1.094 (0.082)
rf	0.324 (0.135)	0.503 (0.077)	0.653 (0.056)	0.766 (0.055)	0.868 (0.036)	1.004 (0.022)
et	0.348 (0.126)	0.52 (0.083)	0.652 (0.054)	0.762 (0.053)	0.861 (0.037)	0.997 (0.02)
gb	0.528 (0.204)	0.884 (0.117)	1.168 (0.081)	1.347 (0.058)	1.451 (0.037)	1.51 (0.021)
hgb	0.666 (0.214)	0.976 (0.116)	1.166 (0.087)	1.286 (0.06)	1.478 (0.035)	1.666 (0.019)
ab	0.407 (0.168)	0.631 (0.104)	0.844 (0.07)	0.971 (0.054)	1.059 (0.04)	1.162 (0.034)
bag	0.326 (0.151)	0.509 (0.095)	0.651 (0.049)	0.767 (0.064)	0.872 (0.044)	1.008 (0.028)
mlp	0.456 (0.095)	0.791 (0.08)	1.246 (0.07)	1.608 (0.049)	1.818 (0.038)	1.962 (0.025)
svr	0.093 (0.026)	0.237 (0.03)	0.465 (0.036)	0.83 (0.049)	1.252 (0.041)	1.626 (0.025)
knn	0.304 (0.23)	0.36 (0.16)	0.378 (0.098)	0.391 (0.073)	0.422 (0.062)	0.464 (0.041)
SuperLearner	1.198 (0.279)	1.719 (0.165)	1.924 (0.09)	1.962 (0.043)	1.977 (0.032)	1.996 (0.02)

Irrelevant variables

Model	N=100	N=250	N=500	N=1000	N=2000	N=5000
lr	0.013 (0.027)	0.004 (0.004)	0.001 (0.001)	0.001 (0.0)	0.0 (0.0)	0.0 (0.0)
lasso	0.0 (0.001)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)
dt	0.013 (0.021)	0.005 (0.017)	-0.002 (0.011)	0.001 (0.008)	0.0 (0.004)	0.002 (0.003)
rf	0.003 (0.005)	0.001 (0.002)	0.002 (0.001)	0.001 (0.001)	0.001 (0.0)	0.001 (0.0)
et	0.002 (0.005)	0.001 (0.002)	0.001 (0.001)	0.001 (0.001)	0.001 (0.0)	0.001 (0.0)
gb	0.004 (0.007)	0.001 (0.002)	0.001 (0.001)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)
hgb	0.002 (0.006)	0.001 (0.002)	0.001 (0.001)	0.001 (0.001)	0.0 (0.0)	0.0 (0.0)
ab	0.004 (0.012)	0.001 (0.005)	0.001 (0.002)	0.0 (0.001)	0.0 (0.0)	0.0 (0.0)
bag	0.004 (0.01)	0.001 (0.005)	0.0 (0.003)	0.001 (0.002)	0.001 (0.001)	0.001 (0.001)
mlp	0.005 (0.018)	0.007 (0.009)	0.005 (0.005)	0.001 (0.003)	0.001 (0.002)	0.003 (0.002)
svr	0.002 (0.002)	0.003 (0.002)	0.005 (0.002)	0.005 (0.001)	0.002 (0.001)	0.0 (0.0)
knn	0.041 (0.184)	0.029 (0.143)	0.012 (0.087)	-0.006 (0.059)	-0.004 (0.043)	-0.006 (0.03)
SuperLearner	0.01 (0.033)	0.004 (0.006)	0.001 (0.003)	0.0 (0.001)	0.0 (0.0)	0.0 (0.0)